

# Complete relations with torsion

## Actual elliptic subgroup indices and the full Pell-7 family

Injective finite torsion coordinates and complete integer kernels remove the free-only index restriction. Strict inverse-unit descent gives every Pell-7 solution and its unique address.

**56**

independent coordinate models

**7,000**

exact relation checks

**2**

complete Pell seed orbits

Witnessed subgroup coordinates, exact finite torsion injection and Smith replay; all-solutions Pell paper proof. No new Lean theorem or global Mordell rank claim.

Source, corpus receipts and reproduction commands accompany the repository snapshot.

## Results and scope

This push closes two bounded arithmetic tasks. Complete prime-preimage packets now admit actual subgroup indices even when rational prime torsion is present. They also admit complete integer relation lattices and reduced invariant factors for their actual source and preimage groups. The Pell-type example  $(x^2+7)/2=y^2$  has a complete, unique two-orbit parametrization, with exact evaluation and inversion. Both results have paper proofs and independently replayable exact computational certificates. This push adds no Lean theorem and changes no audited Lean declaration count.

The index theorem uses a witnessed coordinate model, not a computed full Mordell-Weil basis. It handles repeated and dependent source generators and the entire rational prime kernel, which the earlier square free-coordinate index operation omitted. The model is a certified free subgroup together with an explicitly certified finite torsion subgroup. Every source and every generator of the complete prime preimage must have exact coordinates there. A negative bounded coordinate search is never accepted as evidence of nonmembership. If coordinates cannot be supplied, the existing complete preimage result remains available without a claimed actual index.

## Injective free-plus-torsion coordinates

Let  $E$  be a nonsingular rational elliptic curve. Let  $B_1, \dots, B_r$  be independent modulo torsion, as certified by the existing exact good-reduction independence checker. Let  $T_1, \dots, T_t$  satisfy  $n_i T_i = 0$ . Enumerate every tuple in the finite product of residue classes and require that the corresponding elliptic points are all distinct. This gives an injective homomorphism from the indicated product of cyclic groups to  $E(\mathbb{Q})$ , with the listed orders exact rather than merely annihilating orders.

The combined coordinate map is injective. If a linear combination of the  $B_i$  equals a torsion combination, independence modulo torsion forces every free coefficient to vanish. Injectivity of the finite product then identifies its kernel exactly with the diagonal order relations. It is unnecessary to classify every rational torsion point of the curve: the supplied finite subgroup need only contain the points represented in the packet, including all generators of its prime kernel.

Write the coordinate group and its relation matrix as

$$A = \mathbb{Z}^r \oplus \prod_{i=1}^t \mathbb{Z}/n_i \mathbb{Z}, \quad D = \text{diag}(n_1, \dots, n_t).$$

The implementation accepts zero through four free witnesses, zero through two torsion witnesses, orders at most 32, and a finite product of size at most 256. At least one coordinate direction must be present. Input coordinates are literal integers of absolute value at most 256. These are resource bounds of the executable operation, not restrictions on the abstract theorem or claims about rational torsion classification.

## Complete relation lattices

For a point tuple  $P_1, \dots, P_m$ , let  $C$  be its coordinate matrix, with coordinate vectors as columns. Let  $R$  have  $r$  zero rows above the diagonal torsion matrix  $D$ . Adjoin these torsion order relations as columns:

$$M = [C \mid -R].$$

A complete unimodular Smith transcript certifies  $UMV=S$ . Its columns in  $V$  beyond the rank of  $M$  form a basis of the entire integer kernel of  $M$ . A relation among the elliptic points is exactly a vector  $z$  for which  $Cz=Rw$  for some integer  $w$ . Thus it is exactly the projection onto the first  $m$  coordinates of an element of this kernel.

That projection is injective on the kernel: a vector projecting to zero would satisfy  $Rw=0$ , hence  $w=0$ . Therefore the projected zero-diagonal columns are a complete integer basis of the relation lattice, not a collection of relations detected within a coefficient box. The checker verifies the original augmented matrix, replays every elementary operation, reconstructs these columns and rejects a missing, altered or extra relation basis.

A second Smith transcript reduces the projected relation matrix. It gives the actual invariant factors of the group generated by the point tuple. Factors equal to one are omitted; factors greater than one describe its finite torsion summands, and the free rank is  $r$ . This explicitly distinguishes relations of formal coefficient vectors from relations of the actual rational points.

## Actual subgroup index theorem

Let  $\Gamma$  be the subgroup generated by the source points, and let  $\Gamma'$  be its complete rational prime preimage, already certified by the existing projective-line fibre packet. Coordinates are required for every point in the packet's full generator list, including the rational prime kernel. The earlier replacement-point list alone is not sufficient when torsion is present.

For each group, form the integer lattice generated by its coordinate columns together with  $R$ . Require both lattices to span the whole free witness space. Their rank is then  $r+t$ . Their indices in the ambient integer coordinate lattice are the products of their nonzero Smith factors. The existing preimage replay proves containment; the new checker also verifies source containment by generator-lattice Smith divisibility.

If the two ambient lattice indices are  $a$  and  $b$ , respectively, the exact index is

$$[\Gamma' : \Gamma] = a/b.$$

The proof is the index formula for nested finite-index lattices. Both contain the same torsion relation lattice, so passing to the coordinate group preserves the quotient. The coordinate map is injective, so passing to the actual elliptic groups also preserves it. This works with pure torsion, mixed free and torsion coordinates, and redundant source generators. It does not require that the rational prime kernel be trivial.

On a curve with rational cyclic 5-torsion, the preimage of the zero subgroup has index 5. The corresponding 7-torsion example has index 7. A curve with two independent rational 2-torsion points gives index 4. On  $y^2=x^3-2x$ , with  $P=(2,2)$  and  $T=(0,0)$ , the source  $(2P,T,T)$  gives index 2, while  $(T,T)$  is already closed and gives index 1. The sources  $(5P,T)$  and  $(7P,T)$  give indices 5 and 7. The repeated torsion generator is a deliberate test of complete relations rather than a square determinant shortcut.

## The full Pell-7 solution set

The equation is  $x^2-2y^2=-7$ . Neither coordinate can vanish. Independent signs therefore reduce it to  $x>0, y>0$ . The positive norm-one unit  $3+2\sqrt{2}$  acts by

$$(x, y) \mapsto (3x + 4y, 2x + 3y).$$

The two positive seeds are (1,2) and (5,4). For each seed, apply the displayed transformation any nonnegative number of times; give each resulting coordinate either sign independently. This generates every integer solution, uniquely by its seed, iteration count and two signs. Equivalently, the positive solutions are the coefficient pairs of

$$(1 + 2\sqrt{2})(3 + 2\sqrt{2})^n \quad \text{or} \quad (5 + 4\sqrt{2})(3 + 2\sqrt{2})^n, \quad n \geq 0.$$

The coordinates in either orbit satisfy  $a_{n+2} = 6a_{n+1} - a_n$ . The first orbit begins (1,2),(11,8),(65,46),(379,268); the second begins (5,4),(31,22),(181,128),(1055,746). These are two orbits in the positive quadrant. Combining arbitrary signed seeds with arbitrary integer exponents would obscure the unique positive-quadrant address.

## Completeness and uniqueness by descent

The inverse transformation is  $(x',y')=(3x-4y,3y-2x)$ . It preserves the norm. Since  $x^2=2y^2-7<2y^2$ , one has  $2x<3y$ , and therefore  $y'>0$ . Whenever  $x'>0$ , one also has  $x'<x$ , because  $x<2y$ . Repeated inverse steps must terminate by strict descent on the positive integer  $x$ .

At the terminal point,  $3x\leq 4y$ . Squaring this inequality and substituting the norm equation gives  $2y^2\leq 63$ , hence  $y\leq 5$ . Enumerating  $0\leq y\leq 5$  and testing whether  $2y^2-7$  is a positive square leaves precisely (1,2) and (5,4). Thus every positive solution descends to a listed seed. Conversely, both seeds solve the equation, and forward unit multiplication preserves positivity and the norm.

The inverse is deterministic and the forward transformation is injective. At a seed its inverse leaves the positive quadrant, whereas at any later orbit point its inverse remains positive. Therefore two different seeds or iteration counts cannot give the same positive solution. The sign choices are unique because neither coordinate is zero. This is an all-solutions proof, not an extrapolation from a bounded census. It needs no factorization theorem for quadratic orders and no unproved fundamental-unit assumption.

The finite certificate records the unit identity and every row of the terminal square-root table. Its separate checker validates each floor-square-root interval and reconstructs the seed list, without invoking the producer. orbit uses binary multiplication in the quadratic integer ring. address implements exact inverse descent and raises on step-budget exhaustion. A non-solution returns no address; budget failure is a separate error. Executable orbit indices are capped at 100,000, while the mathematical family permits every nonnegative integer.

## Validation and integration

`make check-torsion-index-pell7` runs the focused test suite, regenerates worked receipts and runs an independent SymPy cross-check. The latter compares 56 finite coordinate models against gcds of every maximal minor, reduces their actual relation lattices independently, and checks 7,000 small coefficient vectors against exact relation membership. These abstract coordinate checks exercise the integral-lattice mechanism; the elliptic worked packets separately exercise actual rational group equations and complete prime fibres.

The Pell census scans all  $0\leq x\leq 100,000$  with integer square roots and compares its full signed output against the orbit generator: 52 signed solutions agree exactly. Large address checks include iteration 1,000 in both seed orbits and all signs. The tests corrupt actual indices, torsion orders, finite injections, coordinates, Smith transforms, relations and group invariants. Producer calls are disabled during elliptic certificate replay. The existing elliptic suite is also rerun to protect historical packets and the free-only operation.

The query service exposes `elliptic_torsion_subgroup_index` and `verify_elliptic_torsion_subgroup_index`, alongside `pell7_complete`, `verify_pell7_complete` and `pell7_address`. The historical `elliptic_subgroup_index` operation retains its original semantics. All new packets retain `execution_verified=false` and do not claim a complete Mordell-Weil group. Independence, rational fibre completeness, finite torsion injection and integer Smith replay are explicit components of the trust boundary.

## Remaining frontier and handoff

The torsion-aware operation closes the previous rational-prime-torsion restriction when a common injective coordinate model and full-rank coordinates are supplied. It does not discover a global free basis or the entire rational torsion group. Automatic coordinate extraction, coordinate models for proper free subspaces, rational division beyond 7, height-based termination, and 2-descent rank upper bounds remain separate tasks. No conditional Mordell census row is relabelled by this work.

The next substantive arithmetic push remains 2-descent rank bounds for selected unresolved Mordell curves. A smaller implementation follow-up is to attach the new torsion-aware index packets to every saturation stage for which the same coordinate model is available. That attachment must preserve the existing rule: failure of a bounded coordinate or membership search gives no negative mathematical conclusion. Formal interpreter refinement likewise remains open until an actual Lean theorem connects typed packets to point and lattice semantics.